Transform Yourself

STUDENT PERSONALITY DEVELOPMENT (SPD)

NAME : VARUNIKA E B B
ROLL NUMBER : 23ADR183
CLASS : AI & DS - ‘C’

TOPIC : AI IN HEALTHCARE

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Estd : 1984

Introduction to Artificial Intelligence

- AI is technology that enables computers and machines to simulate human intelligence and problem-solving capabilities.
- It is ability of machines that think and learn like humans.
- AI systems work by ingesting large amounts of labeled training data, analyzing that data for correlations and patterns.
- By using these patterns it makes predictions about future states.
- AI can perform tasks more efficiently and accurately than humans.
- AI systems and automation tools reduce the time required for data processing.

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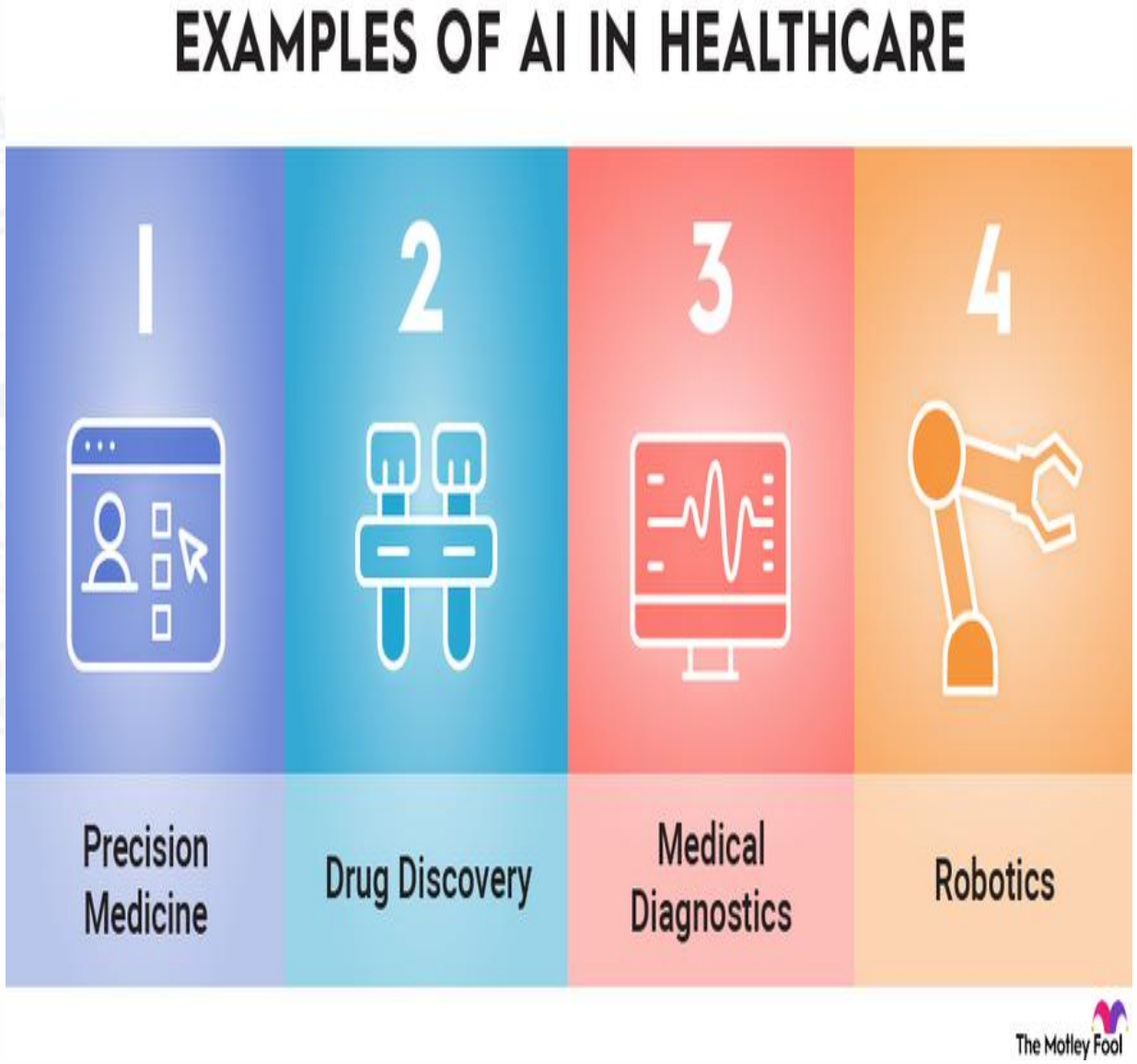
What Is AI in Healthcare?

- Uses machine learning, natural language processing, deep learning to enhance the experience of patients.
- AI analyze relationships between clinical data and patient outcomes.



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Artificial Intelligence In HealthCare

In HealthCare AI is used in the following feilds:

- ✓ Improving medical diagnosis
- ✓ Speeding up drug discovery
- ✓ Transforming patient experience
- ✓ Managing healthcare data
- ✓ Performing robotic surgery
- ✓Administrative tasks

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AI in Medical Diagnosis

- AI analyzes medical images for diagnosis.
- Detects patterns unseen by human eyes.
- Enhances accuracy in disease detection.
- Reduces diagnostic errors and misdiagnosis.
- Speeds up the diagnosis process significantly.
- Improves early detection of serious diseases.

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AI in Drug Discovery

- AI accelerates drug discovery process significantly.
- Analyzes large datasets to identify potential drugs.
- Reduces time and cost in research.
- Identifies new uses for existing drugs.
- Assists in designing drugs with desired properties.
- Enhances precision in targeting specific diseases.
- Integrates data from various research sources.

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AI in Patient Experience

- AI provides personalized healthcare recommendations to patients.
- AI analyzes feedback to improve patient services.
- Provides real-time updates on treatment progress.
- Reduces waiting times with predictive scheduling.

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AI in Healthcare Data Management

- AI speeds up healthcare data processing.
- Reduces human errors in data entry.
- Automates data entry and reduces errors.
- Enhances data security and privacy.
- AI organizes and stores patient data efficiently.
- Speeds up insurance claimming process.

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AI in Robotic Surgery

- AI improves precision in robotic surgery.
- Enhances control and stability of surgical robots.
- Provides real-time data and feedback.
- Reduces surgical errors and complications.
- Supports minimally invasive surgical techniques.
- Reduces recovery time and patient risk.
- Enables remote surgery through robotic control.

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AI in Administrative Task

- AI optimizes appointment scheduling.
- Automates billing and reduces errors.
- Manages patient records efficiently.
- Reduces administrative burden on healthcare staff.

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Pros and Cons of AI in Healthcare

Advantages

- Speeds up drug discovery processes.
- Reduces human error in patient care.
- Reduces costs through operational efficiency.

Disadvantages

- Data privacy and security concerns.
- Risk of errors if AI systems malfunction.
- Ethical concerns about AI replacing human jobs.

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THANK YOU